

Route decision: Shanghai Global Distribution Center → Rotterdam Distripark

HUB-SHANGHAI → HUB-ROTTERDAM · mode any (STRICT) · cargo general · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC · cargo value S\$6,388,320 at 25%/yr carrying cost

Recommendation: BALANCED, SEA via Malacca, Babel, Suez. Expected arrival in **580 h** (24.2 days); plan for 594 h (p85), worst case 614 h (p95). Cost: freight S\$4,981, landed S\$110,779 incl. S\$105,797 carrying cost. Peak leg risk 8%.

Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Landed	Risk
FASTEST	AIR	direct	25	40	46	49	S\$177,301	S\$184,539	2%
SAFEST	AIR	direct	29	42	46	49	S\$178,088	S\$185,727	2%
BALANCED	SEA	Malacca, Babel, Suez	479	580	594	614	S\$4,981	S\$110,779	8%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 39.7 h (p85 45.6 h, p95 49.3 h), S\$184,539 landed, 2 handoffs. 2.2 h faster than the next option. S\$73,761 more (landed) than BALANCED. Avoids Babel, Malacca, Suez.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	S\$45.96	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	S\$191.65	2%	
3	AIR	Frankfurt Airport	8,858	11.1	+8.8	S\$169,763	0%	
4	AIR	Amsterdam Airport Cargo	366	0.5	+8.8	S\$7,010	0%	
5	handoff	Amsterdam Airport Cargo		6.0	+0.0	S\$191.65	2%	
6	ROAD	Rotterdam Distripark	64	0.9	+3.6	S\$98.16	0%	

Transit 12.9 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 24.9 h. Model delay buffer p50/p85/p95: +14.7 / +24.9 / +32.0 h. Route band p50 39.7 h, p85 45.6 h, p95 49.3 h (independent legs); 56.9 h if all legs are late together. Costs: linehaul S\$176,918, transfers S\$383.30, disruption surcharge S\$0.00. Inventory carrying cost S\$7,238 expected (S\$8,988 at p95); landed S\$184,539.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.3 h, p95 49.3 h), S\$185,727 landed, 2 handoffs. 2.2 h slower than FASTEST. S\$74,948 more (landed) than BALANCED. Avoids Babel, Malacca, Suez.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	S\$45.96	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	S\$191.65	2%	
3	AIR	London Heathrow Airport	9,241	11.6	+8.8	S\$177,107	0%	
4	handoff	London Heathrow Airport		6.0	+0.0	S\$191.65	2%	
5	ROAD	Calais Channel Tunnel Hub	169	2.5	+3.6	S\$259.31	0%	
6	ROAD	Rotterdam Distripark	191	2.8	+3.6	S\$292.92	0%	

Transit 17.3 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.3 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.3 h, p95 49.3 h (independent legs); 54.6 h if all legs are late together. Costs: linehaul S\$177,705, transfers S\$383.30, disruption surcharge S\$0.00. Inventory carrying cost S\$7,639 expected (S\$8,988 at p95); landed S\$185,727.

Why the slowest leg (AIR-SHANGHAI → AIR-HEATHROW, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



BALANCED: SEA via Malacca, Babel, Suez

SEA via Malacca, Babel, Suez; expected 580.3 h (p85 594.3 h, p95 613.8 h), S\$110,779 landed, 3 handoffs. 540.6 h slower than FASTEST. S\$73,761 cheaper (landed) than the lowest-cost alternative.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Port of Shanghai	4	0.1	+3.6	S\$6.45	0%	
2	handoff	Port of Shanghai		14.0	+0.0	S\$319.42	8%	
3	SEA	Strait of Malacca	3,810	108.9	+26.6	S\$730.22	0%	
4	SEA	Bab el-Mandeb Strait	6,585	188.1	+27.5	S\$1,262	0%	
5	SEA	Suez Canal	2,302	65.8	+28.5	S\$441.22	0%	
6	SEA	Port of Genoa	2,552	72.9	+20.6	S\$489.18	0%	
7	handoff	Port of Genoa		8.0	+0.0	S\$229.98	5%	
8	RAIL	Lyon Rail Freight Hub	356	5.9	+15.6	S\$227.23	0%	
9	handoff	Lyon Rail Freight Hub		4.0	+0.0	S\$127.77	3%	
10	ROAD	Paris Charles de Gaulle	403	5.9	+3.6	S\$618.24	0%	
11	ROAD	Lille Logistics Hub	184	2.7	+3.6	S\$281.72	0%	
12	ROAD	Rotterdam Distripark	162	2.4	+3.6	S\$247.91	0%	

Transit 452.7 h + transfers 26.0 h + announced scenario delay 0.0 h = plan 478.7 h. Model delay buffer p50/p85/p95: +101.6 / +133.4 / +179.3 h. Route band p50 580.3 h, p85 594.3 h, p95 613.8 h (independent legs); 658.0 h if all legs are late together. Costs: linehaul S\$4,304, transfers S\$677.16, disruption surcharge S\$0.00. Inventory carrying cost S\$105,797 expected (S\$111,905 at p95); landed S\$110,779.

Why the slowest leg (CHOKE-BABEL → CHOKE-SUEZ, p85 28.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.